

VPS & Deployment Infrastructure Provider Comparison (TIAV Deployment)

Scope: Evaluation of VPS, Cloud Compute, and PaaS hosting options for deploying the TIAV FastAPI backend and frontend.
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TIAV Deployment Architecture

The TIAV application offloads LLM processing to OpenRouter, separating the application compute layer from the heavy model inference layer:



Key Architectural Insight: Since LLM inference is managed externally by **OpenRouter**, the TIAV FastAPI backend is primarily responsible for light routing, orchestrating requests, database queries, and serving the frontend. **Dedicated local GPU compute is NOT required** for standard production, making high-performance CPU hosting the most practical and cost-effective approach.

Deployment Provider Overview

PROVIDER	HOSTING TYPE	CPU/GPU	EASE OF SETUP	APPROX COST	SCALING	BEST USE CASE	NOTES
Hetzner	Unmanaged VPS / Dedicated	CPU-Optimized (GPU on dedicated only)	Medium (Manual Linux / Docker setup)	Very Low (~\$4 - \$30/mo)	Manual (Vertical resize or manual load balancer)	Cheapest & High-Performance production hosting	Unmatched price-to-performance ratio; best raw compute value.
DigitalOcean	Managed VPS (Droplets) / PaaS	CPU-Only	Easy (Intuitive UI, 1-click Docker Droplets)	Low-Medium (~\$6 - \$40/mo)	Moderate (Simple horizontal/vertical scaling)	Production-Ready startups and independent devs	Excellent developer portal, managed databases, and highly reliable SLA.
AWS EC2	IaaS VPS	Flexible CPU & GPU instances	Hard (Complex networking, IAM, VPC setup)	Medium-High (\$15+/mo base + egress/EBS)	Excellent (Auto Scaling Groups, ELB integration)	Enterprise deployments with deep AWS integrations	Extremely reliable, industry-standard, but complex billing and setup.
Azure VM	IaaS VPS	Flexible CPU & GPU instances	Hard (Enterprise console, complex IAM/vNets)	Medium-High (Competitive with AWS)	Excellent (Virtual Machine Scale Sets)	Enterprise corporate environments with MS contracts	Seamless Active Directory/SSO integration. Complex pricing structure.
GCP Compute Engine	IaaS VPS	Flexible CPU & GPU instances	Medium-Hard (VPC, IAM, gcloud CLI)	Medium-High (Sustained use discounts)	Excellent (Managed Instance Groups)	Enterprise leveraging GCP's big data/analytics	High-performance internal networking; easy migration to Cloud Run.
RunPod	Serverless GPU / GPU VPS (Pods)	Dedicated GPU + CPU	Easy-Medium (Docker-based templates)	High (~\$100+/mo for persistent VMs)	Easy (Serverless GPU endpoints / multi-pod)	Custom model fine-tuning or secondary local LLMs	GPU-Heavy but redundant for TIAV due to OpenRouter integration.
Vast.ai	Peer-to-peer rented GPU hosting	High-end Consumer & Enterprise GPUs	Medium (Docker, SSH/Jupyter)	Low for GPU (~\$0.10 - \$1.50/hr)	Poor (No native auto-scaler, nodes can go offline)	Non-critical batch jobs or temporary model tests	No SLA/uptime guarantees. Highly discouraged for persistent FastAPI.
Paperspace	Cloud GPU / Core Virtual Machines	Dedicated CPU VMs & Powerful GPUs	Easy-Medium (Web console, Docker templates)	Medium (From ~\$0.07/hr CPU to \$1.50+/hr GPU)	Easy (Simple API scaling, dedicated VMs)	GPU-Heavy with permanent storage & SLA	Reliable alternative to RunPod/Vast.ai with persistent storage.

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Railway	Developer Platform (PaaS)	CPU-Only	Very Easy (Git push deploy, auto FastAPI)	Low-Medium (~\$5 base + usage)	Automatic (Simple horizontal scaling)	Easiest rapid MVP prototyping and deployment	Incredible developer experience. Auto SSL, zero-config CI/CD, and DBs.
Render	Developer Platform (PaaS)	CPU-Only	Very Easy (Git-integrated deployment)	Low-Medium (~\$7/mo for Web Services)	Easy (Autoscaling slider / configuration)	Easiest persistent production web app hosting	Very user-friendly. Built-in DDoS protection, auto SSL, zero-config.

Quick Recommendations for TIAV Deployment

-  **Cheapest (Best Value CPU): Hetzner**
 - *Why:* Offers unmatched CPU and RAM specifications for the price, making it ideal for persistent, low-budget, high-throughput production hosting.
-  **Easiest (Best Developer Experience): Railway or Render**
 - *Why:* Zero-config deployment directly from your Git repository. Handles automatic SSL certificates, database hosting, and CI/CD out-of-the-box.
-  **Enterprise-Grade (Complex & Highly Scalable): AWS EC2, GCP Compute Engine, or Azure VM**
 - *Why:* Perfect if your organization requires rigid enterprise security policies, complex VPC setups, or needs to hook directly into active cloud infrastructures.
-  **GPU-Heavy (Alternative/Secondary Local Inference): Paperspace or RunPod**
 - *Why:* If you decide to host a secondary lightweight local model (like Nemotron-Mini) alongside OpenRouter, Paperspace provides dedicated SLA-backed persistent GPU VMs.
-  **Production-Ready (Balanced Compute & DevOps ease): DigitalOcean**
 - *Why:* The ideal middle ground between unmanaged VPS and complex cloud platforms. Offers reliable SLAs, managed databases, and basic Kubernetes setups.